

CSCI 331 (Section **54**) [001420]
Software Systems
Spring 2026 SYLLABUS

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 please prefix "**CSCI331:** " in the Subject field)

Web Site: <https://aanda-stcloudstate-edu.github.io/WebSpace/cs331.html>

Instructor Schedule:

<https://aanda-stcloudstate-edu.github.io/WebSpace/Wkly-sched-w26.pdf>
(and by appointment)

Student Support Hours: <https://minnstate.zoom.us/j/91845917984>

{I sometimes won't be available for a Student Support hour -
this will usually be because I'm attending a meeting}

Resource Links:

- [D2L Brightspace](#)
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- [SCSU Student Resources for Online Learning](#)
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- [SCSU Zoom Resources](#)
-
- [SCSU Student Information for Attending Classes Off-Campus](#)
-
- [SCSU: The Write Place](#)
-
- [SCSU Information Technology Services](#)
-
- [SCSU Medical Clinic](#)
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- [SCSU Counseling and Psychological Services](#)
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- [SCSU Library](#)
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- [SCSU Student Code of Conduct](#)

Additional SCSU student resources, curated and compiled by the Academic Affairs office, are presented in:

[Fall 2023 Student Instructional Resource and Support Guide](#)

Hyflex Content Delivery and Accessibility:

I intend to record all lectures. And, most materials will be available through D2L.

Student submissions of quizzes, assignments, notes, etc. will be via D2L.

CLASS TIME AND LOCATION:

Class: **T R: 14:00 - 15:15 in Completely Online Synchronous** Section **54**

Lecture Zoom Meeting ID: <https://minnstate.zoom.us/j/96045763773>

REQUIRED TEXT: * Data Abstraction & Problem Solving with C++,
Carrano, Pearson, 6th, 7th, or 8th ed.

SUPPLEMENTARY TEXTS: * File Structures: An Object-Oriented Approach with
C++, M.J. Folk, B. Zoellick, G. Riccardi, Addison Wesley Longman, 1998.

[A PDF copy of this text (which is out-of-print) accessible is on our course
D2L page in Materials/Content/Primary Textbook: FZR. (Please use this copy
only for this course)]

* Applied Data Structures with C++, Peter Smith, Jones
and Bartlett, 2004.

* your CSCI 201 C++ textbook

COURSE DESCRIPTION:

Problem solving strategies and concepts applied in the context of issues
associated with the design and implementation of software systems using
a combination of current software packages/environments. Subjects
addressed include file processing, data modeling and mapping to storage
structures, data base systems, and software design and implementation.

STUDENT LEARNING GOALS:

1. gain experience working in a group in various roles on a technical project.
2. gain knowledge updating sequential files as related to data base systems.
3. gain experience designing software systems through a software engineering process including designing algorithms, coding, and testing.
4. write programs and software systems involving trees, blocking/deblocking, sort/merge, updating sequential files, and B+trees.

I Student Outcomes as defined by ABET (CAC):

- a) An ability to apply knowledge of computing and mathematics appropriate to the discipline.
- c) An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.
- d) An ability to function effectively on teams to accomplish a common goal.
- e) An understanding of professional, ethical, legal, security and social issues and responsibilities.
- f) An ability to communicate effectively with a range of audiences.
- i) An ability to use current techniques, skills, and tools necessary for computing practice.
- j) An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems
in a way that demonstrates comprehension of the tradeoffs involved in design choices.
- k) An ability to apply design and development principles in the construction of software systems of varying complexity.

II Student Outcomes (Sos) for the Computer Science Program (applicable to this course):

Students will be able to:

1. apply structured principles and good practices to the task of developing software systems.

4. communicate both technical and non-technical aspects of their work in formal and informal situations.

5. apply formal methods to the process of constructing a system and an appreciation of the need to study and develop such methods.

7. analyze various aspects of the process used when designing a system and employ established frameworks to evaluate the completed work.

TOPICS (not necessarily in order of coverage):

- * Sequential file processing
- * Indexed Files
- * Hashed Files
- * Tree Structured Files
- * Secondary storage media
- * Introduction to database
- * Software Design Processes and Principles
- * Software Development Processes and Principles
- * and more...
- * (I will expect students to learn the complexities $\{O(n)\}$ of the operations on the data structures we discuss and to be able to select, based on those complexities, the most appropriate ADT considering the nature of the data and how it will be accessed.)

SLOs:

* Apply structured principles and good practices to the task of developing software systems.

* Communicate both technical and non-technical aspects of their work in formal and informal situations.

* Apply common formal methods to the process of constructing a system and an appreciation of the need to study and develop such methods.

* Analyze various aspects of the process used when designing a system and employ established frameworks to evaluate the completed work.

PREREQUISITES:

[Prereq.: CSCI 220 or ECE 221; CSCI 301 (301 coreq: Math 271)]

I assume that all students in this class have passed the prerequisite CSCI and MATH courses, or their equivalents.

We also assume that you have an operational facility with Unix.

If you have not had ALL of these courses you should drop this course until you have passed all of the prerequisites.

We do not have time for an extensive recapitulation of topics covered in the prerequisite courses.

PROGRAMMING ASSIGNMENTS HOMEWORK AND IN CLASS WORK:

There will be a sequence of programming assignments.

These must be submitted by the time and date as indicated on the respective D2L DropBox.

No allowances will be made for computer down time, so I suggest you begin each assignment as soon as you receive it.

Programs not submitted by their deadline will be penalized.

A program must compile to receive any credit.

(Which implies that a successful strategy is to always maintain available the

last successfully compiling version as you are developing incrementally)

Keep in mind that I expect all programs to produce correct output given error-free input.

Check your output before submitting your final copy (The grader will ignore all but your final submission).

Don't assume that a program which produces any output, without terminating abnormally, is correct.

Are the output and test results what you expected?

Test your program carefully and extensively.

You have not fully tested your program if you have not executed every line of code.

Preconditions and postconditions must be consistent and well-defined.

The code should satisfy the postconditions if the preconditions are met.

Conversely, if a precondition is violated, that exception should be detected and handled.

The grade you get on a program is based on the code, the documentation and how well it is tested. To receive better than a C on a program it must do

more than produce correct output given correct input. By this I mean your

program must be robust, well documented, well tested and well written.

Your grade on the group projects will also be based on how well you cooperate with, contribute to, and participate in the group you are assigned to. Your contributions to your group's efforts will be assessed by your peers.

* "Boat anchors" will be significantly penalized up to a whole letter grade.

* "Heroes" will be significantly rewarded -- up to a whole letter grade.

Missing class on days that we have in-class assignments or quiz will result in a

zero for that assignment. These assignments cannot be made up. If you have trouble getting to class drop now so that someone else can use the spot. I will be dropping the lowest quiz score from the quiz score total.

ACADEMIC HONESTY:

You are expected to do your own homework. If you copy someone else's work or allow someone else to copy your work, you are being academically dishonest and will be subject to severe disciplinary action which may include any or all of: no credit for the work in question, a failing grade

for the course, notification to the university that you have violated your

Code of Conduct. Use of recording devices during exams is prohibited.

If you must quote or paraphrase another source, citation is essential, otherwise plagiarism has been committed. You are expected to be familiar with your rights and obligations as outlined in the

[Student Code of Conduct](#) - specifically the [Prohibited Student Conduct](#) section, and the [Academic Integrity policy](#) within.

STUDYING IN GROUPS:

I encourage you to study with someone else in the class, but when you prepare the final documents to turn in be sure that it is your own work and that you understand it.

If you represent someone else's work as your own (without citing them as a source), or allow someone else to turn your work in as theirs (without citing you as a source), you will have committed academic dishonesty and will receive an F.

ATTENDANCE:

You are responsible for knowing what happens at each class meeting. You should expect to spend a lot of time on this class. It is not an easy class. If you are having trouble in class I expect to see you in my office. If you don't tell me you are having troubles how do you expect me to know? If you can't come during my office hours, make an appointment for another time.

QUIZZES:

There will be a sequence of quizzes of assigned readings (usually prior to that chapter's coverage in class).

EXAMS:

- * There will be one or two midterm exams and a **comprehensive** final exam.
- * Exams will be based on the notes; that is, on the presentations in class as well as on the assigned readings and the assignments.
- * Make-up exams will be given in only in extreme (or university sanctioned) circumstances and only with prior notification to the instructor.

GRADING:

Your grade will be determined by the contributions of your scores on assignments, quizzes, intermediate and final exams. Makeup of tests, quizzes, and exams are by prior arrangement only.

APPROXIMATE POINTS:

Assignments	35%
Sectional Exam(s)	25%
Final Exam	40%

FINAL EXAM: Section 54: **Monday, May 4, 12:20 - 14:35** as a **D2L Quiz**.

Prohibited: use of cell phones, AI, or people other than the instructor.

CAVEAT:

I reserve the right to amend the contents of this syllabus with notification.

AI:

While Artificial Intelligence (AI) is a tool available to the public, within this course, you are expected to submit assignments, summaries, and discussions that represent your original thoughts.

If an assignment is suspected of being AI-generated, there will be additional conversations between the faculty and student, similar to our plagiarism practices.

Please refer to the Academic Integrity policies that apply to all courses within our institution.

EXPLORATORY USE OF AI: (*Supportive but Limited*)

Supported tools such as Copilot and Adobe AI tools may be used in this course for limited academic tasks such as idea generation, outlining, or early-stage revisions.

Use of agentic AI tools integrated transparently into software products such as Grammarly and Overleaf are generally OK.

However, all submitted work must reflect your own understanding, voice, and academic reasoning.

AI tools should serve as a support, not a replacement, for your thinking. Submitting work that is primarily AI-generated without acknowledgment is a breach of our Academic Integrity Policy. If you use AI tools as part of your process, you must be transparent about how they were used (e.g., "Used Copilot to draft bullet points; expanded and revised with personal analysis"). When in doubt, ask for clarification before submitting your work. Please consult the Academic Integrity Policy for more information and let's maintain an open dialogue about ethical and responsible technology use.

SAFETY:

All students are encouraged to critically reflect on the course topics and to raise questions to the class and to the professor. Please be respectful to one another by not presenting your question in a hostile manner.

Open dialogue on course content is encouraged, but attacks on classmates or me are not. Please set up a time to talk with me if you have questions about this policy or if you believe this policy is being violated.

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Student Services

Counseling and Psychological Services

College can be stressful and it is normal for some students to struggle emotionally while balancing the demands of college and personal lives. If you are struggling, please stop by and see me during Student Support Hours (office hours). Know that our Counseling & Psychological Services (CAPS) Department is available Monday through Friday 8-4:30 to assist students who need help. Don't hesitate to reach out for help. We care about you and want you to be successful both personally and professionally here at SCSU.

How to get an appointment or make a referral to CAPS:

1. Call 320-308-3171
2. Go to the Request an Appointment link on the website: <https://www.stcloudstate.edu/counseling/>
3. Walk in to the CAPS clinic Monday through Friday any time between 8am and 4:30pm

4. Faculty/staff can call CAPS or walk over to CAPS in person with a student for a warm hand-off (we are located in Eastman Hall, 3rd floor). Faculty/staff should call ahead when possible so we can be prepared for an immediate visit with the student

After hours use any of these crisis resources:

Local crisis line at 320-253-5555 or 800-635-8008

[National Suicide Prevention Lifeline](#) or call 1-800-273-8255

Crisis Text Line: Text "MN" to 741-741

Food and Housing Insecurity

If you face challenges securing food or housing and believes these needs may affect your academic performance in this course, please contact the Case Manager housed in Counseling and Psychological Services (CAPS) at 320-308-3191 for a consult and possible assistance. Feel free to talk to me if you need assistance in accessing this resource.

For more information on the [Huskies Food Pantry](#) or e-mail the Pantry foodpantry@stcloudstate.edu.

"Keep the Pack Safe" Statement

St. Cloud State University (SCSU), in coordination with state and local health departments, is closely monitoring the spread of COVID-19 and following the State of Minnesota's laws and guidelines to keep everyone safe. As a reminder:

If you are not feeling well, be sure to call the SCSU Medical Clinic for assistance at (320) 308-3191 or email myhealthservices@stcloudstate.edu.

LGBTQIA

The LGBT Resource Center at St. Cloud State University promotes full inclusion of LGBT+ folk and allies and works to dismantle prejudice, discrimination, and oppression toward and within the LGBT+ community. Visit the [LGBT Resource Center](#) for more information regarding the offered support services at St. Cloud State University.

Our Husky Compact

As our student, we make a bond with you in [Our Husky Compact](#) to prepare you to be a global citizen of the 21st Century. As a graduate of St. Cloud State, you will be prepared for a life of intellectual, professional and personal fulfillment with the skills to discover and apply new solutions to challenges and opportunities through risk-taking, innovation and imagination. Lifelong growth and development along the dimensions of Our Husky Compact will prepare all members of the St. Cloud State community to reconsider the world around them.

Sexual and Relationship Violence

St. Cloud State University (SCSU) is committed to fostering a safe and inclusive learning environment for all students. If you (or someone you know) has experienced/survived any form of sexual harassment or sexual violence, including rape, sexual assault, relationship or inter-partner violence, or stalking, know that help and support are available. SCSU strongly encourages all individuals to report incidents of sexual harassment to the Title IX Coordinator:

Judith Siminoe
Interim Title IX Coordinator and Designated Officer
Office of Institutional Equity & Access
121 Administrative Services Building
Phone: 320-308-5123
Email: oea@stcloudstate.edu

Additional information and online reporting options are available [here](#).

Please be aware that if you share, verbally or in writing, incidents of this nature, as your instructor, I am required to disclose this information to Title IX Coordinator, the designated employee to coordinate the university's efforts to comply with Title IX federal and state requirements. If you wish to speak to a confidential employee or resource, who does not have this responsibility, you can contact the following resources:

<u>SCSU Counseling and Psychological Services</u>	320-308-3171
<u>SCSU Medical Clinic</u>	320-308-3191
<u>Women's Center</u>	320-308-4958
Central MN Sexual Assault Center (Community program)	320-251-4357 800-237-5090 24-hour hotline

Your safety and success as an SCSU student are of utmost importance. If you have any questions or want additional information, please contact the Office for Institutional Equity and Access at (320) 308-5123, via email at oea@stcloudstate.edu, or in person.

Student Accessibility Services

St. Cloud State University is an affirmative action, equal opportunity employer and educator. St. Cloud State University is committed to a policy of nondiscrimination in employment and education opportunity and works to provide reasonable accommodations for all persons with disabilities. Accommodations are provided on an individualized, as-needed basis, determined through appropriate documentation of need. Please contact [Student Accessibility Services](#) or 320-308-4080, office Centennial Hall (CH) 202 to meet and discuss reasonable and appropriate accommodations for your plan. The accommodations authorized in your plan should be discussed with your instructor. All discussions will remain confidential. This syllabus is available in alternate formats upon request.

Our Land Commitment

St. Cloud State University is committed to developing a land acknowledgment that is not just words but actions. We will actively promote education among our faculty, staff, and students on Indigenous issues, including research on the historical and ongoing harms impacting our Indigenous communities. We will also clearly outline our goals to make a positive and lasting contribution to efforts to build relationships with the Tribal Nations represented within our local and broader community. St. Cloud State University commits to creating a substantive and empathic land acknowledgment, demonstrating our continued commitment to action. Faculty are encouraged to learn more about **Our Land Commitment** on the [OEA Land Commitment webpage](#) for the most up to date information.

EAB Navigate Student Success Tool

St. Cloud State University uses the Navigate platform, which allows faculty and advisors to provide proactive and coordinated care for students. Throughout the semester, I may submit updates on students with attendance or academic concerns so that advisors can reach out to provide additional support and/or connect you to campus resources. I may also contact you through email, before or after class, or encourage you to meet with me during my Student Support hours. Please know I'm here to support you in your academic success. You can learn more about the Navigate tool at <https://www.stcloudstate.edu/universitycollege/navigate/student-resources.aspx>

Student Support Hours:

The time that I have listed as student support hours is time for you. It is time that I have dedicated to be available for whatever you need. This does not mean that I will stare sadly out the window waiting for a student to arrive.

I will be doing other work. What it does mean is that the second you show up, that work goes away, and my time is yours. We can talk about the course, the department, some other topic, grad school, etc.

I am available for you during that time. Additionally, if my door is open at other times, feel free to drop. To guarantee my availability, please make an appointment through **Navigate**.

I'm in my office often. If I am in but can't talk right then, I'll let you know, then we can chat at a different time.

I am also available via Zoom and telephone during my Student Support Hours.